WOLLO UNIVERSITY

KOMBOLCHA INSTITUTE OF TECHNOLOGY (KIOT)

DEPARTMENT OF INFORMATION TECHNOLOGY

INDUSTRIAL INTERNSHIP REPORT

ON

CAMPUS NETWORK DESIGN, IMPLEMENTATION, AND STRUCTURED CABLING

Hosted at: Wollo University – Kombolcha Institute of Technology (KIoT), ICT and Network Infrastructure Directorate

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Company Supervisor: Kalid Ali (Wollo University – KIoT)

Duration of Internship: 45 days

Date of Submission: September 27, 2026

Wollo University (KIOT), Kombolcha, Ethiopia

DECLARATION

Student's Declaration

I, **Semre Demssie**, a final-year undergraduate student in the Department of Information Technology at Wollo University, hereby declare that this industrial internship report titled "Campus Network Design, Implementation, and Structured Cabling" is an original account of the work performed by me during my 45-day internship placement at Wollo University – Kombolcha Institute of Technology (KIoT) under the ICT and Network Infrastructure Directorate, from July 1 to August 14, 2026.

All primary data, physical cabling procedures, topology designs, and device configurations documented herein were executed by me under supervision. Secondary sources, standards (such as ANSI/TIA-568), and reference materials have been duly cited and acknowledged. This report has not been submitted previously to Wollo University or any other academic institution for the award of any degree or qualification.

Student Name: Semre Demssie

Signature: _____

Date: September 27, 2026

Approval of the Industrial Supervisor (Host Mentor)

This is to certify that Semre Demssie has successfully completed the required 45-day industrial internship training at Wollo University – Kombolcha Institute of Technology (KIoT) under my direct supervision within the ICT and Network Infrastructure Directorate. The student participated actively in physical structured cabling tasks (UTP/STP termination, RJ45 crimping, cable testing, fiber handling) and logical network engineering (VLANs, routing, DHCP, SSH, and firewall access control). This report accurately represents the practical work executed by the student during the placement.

Supervisor Name: Kalid Ali

Title: Network Infrastructure Lead, ICT Directorate, Wollo University (KIoT)

Signature: _____

Date: September 27, 2026

Approval of the Academic Advisor

I hereby confirm that I have evaluated this industrial internship report submitted by Semre Demssie and approve it as fulfilling the documentation requirements for the undergraduate internship course in the Department of Information Technology, Faculty of Technology, Wollo University.

Academic Advisor Name: Muluneh Endalew

Department: Information Technology, Wollo University

Signature: _____

Date: September 27, 2026

ACKNOWLEDGEMENTS

First and foremost, I would like to express my deepest gratitude to Wollo University, specifically the Department of Information Technology and the University-Industry Linkage Office, for granting me the opportunity to undertake this practical industrial internship, which bridged the gap between my classroom theoretical knowledge and real-world network engineering practices.

I am profoundly indebted to my host institution, Wollo University – Kombolcha Institute of Technology (KIoT), and its ICT and Network Infrastructure Directorate, for opening their doors to me and providing access to their active campus networking environment, server rooms, and structured cabling tools.

My sincere thanks go to my Industrial Supervisor, Mr. Kalid Ali, for his continuous mentorship, professional guidance, and patient instruction during hands-on sessions involving physical cable crimping, patch panel punching, and active switch/firewall configurations. His technical insights into campus-wide network administration were invaluable.

I extend special recognition to my Academic Advisor at Wollo University, Mr. Muluneh Endalew, for his guidance, periodic follow-up, constructive feedback, and advice on aligning my technical deliverables with academic guidelines.

Furthermore, I acknowledge the dedicated team of network technicians, systems administrators, and IT support staff at KIoT who generously shared their daily field experience and assisted me during physical cable pulling and testing sessions.

Lastly, I express my deep gratitude to my family and fellow classmates for their unwavering moral support, encouragement, and understanding throughout the internship period.

EXECUTIVE SUMMARY

This industrial internship report documents a 45-day practical attachment completed at Wollo University – Kombolcha Institute of Technology (KIoT) under the ICT and Network Infrastructure Directorate. The overarching objective of the internship was to gain comprehensive, end-to-end practical experience in both physical-layer structured cabling and logical network architecture for a large institutional campus.

The practical training was divided into two main domains: Physical-Layer Cable Engineering and Logical Network Design & Security Implementation. In the physical layer, the intern performed hands-on site preparation, cable cutting, jacket stripping, conductor alignment, and RJ45 crimping following ANSI/TIA-568 standards (T568A and T568B pin-outs), processing and testing both UTP (Cat5e/Cat6) and STP cables, and handling fiber-optic patch cords.

In the logical layer, the intern designed and deployed a scalable , 3-tier hierarchical network architecture (Core–Distribution–Access) serving four primary campus buildings: Ict-Building, Administer, Library, and Student-Access. Configurations executed in Cisco IOS and ASA CLI included VLAN segmentation, Inter-VLAN routing via Layer-3 switch virtual interfaces (SVIs), static and default routing, centralized DHCP allocation, SSH-based management hardening, and extended Access Control Lists (ACLs) to isolate student traffic from administrative subnets.

The completed design was simulated, cross-verified, and validated using Cisco Packet Tracer, matching the physical cabling layout installed on site. Operational challenges — including crimping failures, addressing overlaps, and ACL placement errors — were systematically diagnosed and resolved through standardized checklists. This report follows the institute's internship documentation guideline: it begins with the background of the host institution, moves through the internship experience and benefits gained, presents the full project work (problem, objective, methodology, literature review, and analysis), and closes with conclusions and recommendations. The internship strengthened the intern's technical proficiency, industrial problem-solving capability, teamwork, professional ethics, and technical documentation skills.

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CHAPTER ONE: BACKGROUND OF THE HOST INSTITUTION

1.1 Brief History of Wollo University – Kombolcha Institute of Technology (KIoT)

1.1.1 Overview of Wollo University

Wollo University is one of the public higher education institutions in Ethiopia, established to contribute to the country's socio-economic development through quality education, applied research, and community engagement. The university operates across multiple campuses, with its primary administrative seat located in Dessie and its technology-focused institute situated in Kombolcha. Over the past two decades, Wollo University has expanded its academic programs across various faculties, including engineering, technology, health sciences, agriculture, business, and social sciences.

1.1.2 Evolution of Kombolcha Institute of Technology (KIoT)

Wollo University – Kombolcha Institute of Technology (KIoT) was established as a specialized technology institute under Wollo University to address the growing national demand for qualified engineers, computer scientists, and Information and Communication Technology (ICT) professionals. Located in Kombolcha — a major industrial and manufacturing hub in the Amhara Regional State — KIoT is positioned to foster collaboration between higher education and industry.

Since its founding, KIoT has grown significantly in terms of student enrollment, academic staff, research output, and physical infrastructure. The institute hosts several engineering faculties, including:

- School of Electrical and Computer Engineering
- School of Computing and Software Engineering
- School of Mechanical and Industrial Engineering
- School of Civil and Water Resources Engineering

To support these academic units, KIoT established a dedicated ICT infrastructure connecting academic blocks, administrative offices, research laboratories, libraries, and student access areas.

1.2 Main Products and Services of the ICT Directorate

The ICT and Network Infrastructure Directorate at KIoT serves as the primary technical backbone for the campus. It plans, deploys, operates, and maintains all digital and network services across the institute, organized into four core service lines: LAN/WAN Infrastructure & Structured Cabling, Enterprise Network Security, Data Center & Systems Administration, and IT Support & Maintenance.

1.2.1 Campus Network Infrastructure & Structured Cabling

- Design, expansion, and maintenance of the campus Local Area Network (LAN) and Wide Area Network (WAN) backbones.
- Installation and termination of copper (UTP/STP Cat5e, Cat6) and fiber-optic cabling across campus buildings.
- Management of rack equipment, patch panels, distribution frames, and access switches in all building telecommunication closets.

1.2.2 Enterprise Network Management & Security

- Deployment and maintenance of core routers, distribution switches, edge firewalls (such as Cisco ASA), and wireless access points.
- Network segmentation using VLANs and Inter-VLAN routing to maintain performance and data security.
- Implementation of traffic filtering policies via Access Control Lists (ACLs) to protect administrative subnets.

1.2.3 Data Center Operations & Systems Administration

- Management of central server resources hosting essential academic and administrative applications.
- Centralized network service administration, including DHCP and DNS services.

- Administration of student information systems, institutional e-learning platforms, and staff web portals.

1.2.4 IT Helpdesk & User Support Services

- Providing technical support to staff and students for workstation connectivity, software configuration, and network troubleshooting.
- Maintenance and repair of computer laboratory equipment, printers, and peripheral devices across campus.

1.3 Main Customers and End-Users of the Network

The ICT Directorate provides connectivity to diverse end-user groups across the campus. Each group requires tailored network policies, bandwidth allocations, and access privileges.

1.3.1 Administrative Staff (Administer Building)

Role: university management, human resources, finance, registrar, and campus administration. Network requirement: high-priority, low-latency, secure connectivity with restricted access to administrative servers and financial databases; assigned to dedicated staff subnets (VLAN 20).

1.3.2 Academic Staff and ICT Technical Personnel (Ict-Building)

Role: professors, researchers, laboratory instructors, and network engineering personnel. Network requirement: high-bandwidth access for research data transmission, server management, and academic collaboration; assigned to ICT subnets (VLAN 10).

1.3.3 Students and Library Patrons (Library Building)

Role: undergraduate and postgraduate students accessing digital libraries, online journals, and research repositories. Network requirement: standard web access with content filtering and controlled bandwidth allocation; assigned to Library subnets (VLAN 30).

1.3.4 General Student Body (Student-Access Building & Wi-Fi Zones)

Role: students connecting personal laptops, tablets, and mobile devices in designated study areas and access centers. Network requirement: isolated internet access with strict firewall rules blocking

lateral movement toward staff and administrative subnets; assigned to Student-Access subnets (VLAN 40).

1.4 Organizational Structure of the ICT Directorate

The ICT and Network Infrastructure Directorate operates under a hierarchical structure to ensure administrative oversight and technical accountability. The Directorate reports to the KIoT Administration (Vice Dean / Administrative Director) and is organized into three operational units: the Network Engineering & Structured Cabling Unit (where the intern was placed), the Systems & Server Administration Unit, and the IT Helpdesk & User Support Unit.

1.4.1 Organizational Roles and Functions

- Directorate Office: oversees strategic IT planning, budgeting, technology procurement, and policy formulation.
- Network Engineering & Structured Cabling Unit: responsible for physical cabling installation, switch/router/firewall configuration, network monitoring, and fault repair. The intern was attached to this specific unit during the 45-day placement.
- Systems & Server Administration Unit: manages server operating systems, databases, cloud integration, user accounts, and data backups.
- IT Helpdesk & User Support Unit: handles user support tickets, workstation maintenance, and computer lab operations.

1.5 Operational Workflow within the ICT Directorate

To manage network projects, infrastructure upgrades, and daily maintenance requests, the ICT Directorate follows a structured 8-step operational workflow, moving sequentially from request identification through to documentation and handover:

1. Request / Need Identification
2. Site Survey & Assessment
3. Technical Design & Bill of Quantities (BOQ)
4. Administrative Approval & Procurement

5. Physical Cabling & Hardware Installation

6. Logical Configuration & Policy Enforcement

7. System Verification & Testing

8. Documentation, Handover & Maintenance

Detailed Description of Workflow Steps

1. Request / Need Identification: a department or building submits a service request for new network connections, bandwidth expansion, or infrastructure troubleshooting.

2. Site Survey & Assessment: network engineers visit the physical location to assess building structural features, cable pathway requirements, user port counts, and power availability.

3. Technical Design & BOQ Preparation: engineers draft the network topology, IP addressing plan, and a detailed Bill of Quantities covering cables, patch panels, faceplates, RJ45 connectors, and active switches.

4. Administrative Approval & Procurement: the project design and material requisitions are reviewed and approved by the ICT Director and institute management.

5. Physical Cabling & Hardware Installation: the cabling team pulls UTP/STP/fiber cables through conduit, punches cables down to patch panels, terminates RJ45 jacks, and mounts switches into equipment racks.

6. Logical Configuration & Policy Enforcement: engineers configure switch ports, VLANs, Inter-VLAN routing gateways, DHCP pools, SSH access, and firewall security ACLs.

7. System Verification & Testing: the completed setup undergoes physical wire testing (continuity/wire-map) and logical packet verification (ping, traceroute, DHCP lease check, ACL validation).

8. Documentation, Handover & Maintenance: as-built network diagrams and IP configuration records are archived, and the network is handed over to the requesting department for live production use.

CHAPTER TWO: INTERNSHIP EXPERIENCE

2.1 How the Internship Placement Was Secured

The placement at Wollo University – Kombolcha Institute of Technology (KIoT) was arranged through the Department of Information Technology's University-Industry Linkage Office, which matches final-year students with host organizations offering hands-on, industry-relevant technical work. Wollo University – KIoT was selected because of its modern, enterprise-grade ICT infrastructure. As a specialized technical institute hosting thousands of students and faculty members across multiple faculties, KIoT operates a complex multi-building enterprise campus network featuring high-speed optical fiber backbones, managed Layer-2/Layer-3 switches, perimeter firewalls, and dedicated server rooms.

This environment provided an ideal industrial setting to bridge the gap between theoretical networking models — such as the OSI 7-layer model, IP addressing schemes, and routing protocols — and live, physical deployment, structured cabling, and enterprise network security.

2.2 Section of Assignment and Role

During the 45-day internship, the intern was assigned directly to the Network Engineering & Structured Cabling Unit under the ICT and Network Infrastructure Directorate. Mentored by Senior Network Engineer Mr. Kalid Ali, the intern functioned as an Assistant Network Engineering Trainee. The scope of work encompassed two core operational domains:

- Physical Layer Engineering: on-site structured cabling, cable pulling through conduits, patch panel termination, modular jack punching, UTP/STP RJ45 cable crimping, optical fiber patch cord management, and hardware cable testing.
- Logical Layer Engineering & Security Architecture: campus network topology design, IP address allocation planning (10.172.0.0/16), VLAN segmentation, Inter-VLAN routing, perimeter firewall integration (Cisco ASA), centralized DHCP configuration, SSH management hardening, and extended ACL traffic filtering.

2.3 Work Flow in the Section

The Network Engineering & Structured Cabling Unit follows the same 8-step operational workflow used across the ICT Directorate (Section 1.5). Within that workflow, the intern was directly and continuously involved in three consecutive stages: Physical Cabling & Hardware Installation, Logical Configuration & Policy Enforcement, and System Verification & Testing. Requests were first triaged by the Directorate Office, after which a site survey and technical design were prepared by senior engineers; the intern then joined the team from the physical implementation stage onward, working alongside technicians through to final documentation and handover.

2.4 Work Pieces and Tasks Executed

The technical work executed during the internship covered the complete lifecycle of campus network deployment, organized into four primary technical modules: Physical Structured Cabling & Termination; Topology Architecture & Hardware Interconnection; Layer-2/Layer-3 Switch & Routing Configuration; and Perimeter Firewall & Security Enforcement. A full technical breakdown of each module, including configuration excerpts and IP design tables, is presented in the Project Work chapter (Chapter Four); this section summarizes the tasks at the level of day-to-day duties.

- Physical structured cabling across four primary campus facilities: Ict-Building, Administer, Library, and Student-Access — including cable measuring, jacket stripping, conductor ordering, RJ45 crimping, and patch panel punch-downs.
- Building and interconnecting a scalable 3-tier hierarchical network hardware layout: a Cisco ASA 5505 perimeter firewall, a Cisco Catalyst 3650 core switch, two Cisco Catalyst 3560 distribution switches (D1, D2), and four Cisco Catalyst 2960 access switches.
- Configuring VLANs, Switch Virtual Interfaces (SVIs) for Inter-VLAN routing, static and default routes, and centralized multi-pool DHCP services on the distribution switches.
- Hardening administrative access with SSH version 2 and enforcing extended Access Control Lists (ACLs) to isolate student traffic from administrative and ICT resources.

2.5 Procedures Followed While Performing the Tasks

Each task was carried out according to a standardized procedure rather than ad-hoc execution, consistent with the Directorate's workflow. For physical cabling, this meant measuring and cutting cable to length, stripping the jacket, ordering conductors to the ANSI/TIA-568 T568A or T568B standard, trimming, inserting into an RJ45 connector, crimping, and finally continuity-testing every terminated run before it was accepted into service. For logical configuration, the intern followed a design-first approach: the IP addressing and VLAN plan was finalized on paper before any device was touched, configurations were entered through the Cisco IOS and ASA command-line interfaces in a fixed order (hostname and management access, then VLANs and interfaces, then routing, then DHCP, then security policy), and every change was saved with write memory and verified before moving to the next device. The complete step-by-step procedures, wiring standards, and CLI excerpts are documented in Chapter Four (Project Work).

2.6 Performance in Executing the Tasks

Performance was measured against the same end-to-end verification checklist the Directorate uses to accept newly installed infrastructure into production (see Table 8 and Appendix C). Every physical wiremap test, DHCP lease negotiation, inter-VLAN ping test, ACL enforcement test, perimeter routing trace, and SSH management session initiated during the internship returned a PASSED result, with zero percent packet loss recorded between building gateways and round-trip times under two milliseconds on the internal backbone. The intern's terminated cable runs and CLI configurations were reviewed and accepted by the Industrial Supervisor without requiring rework beyond the specific corrective actions described in Section 2.8, indicating that the tasks were executed to the standard expected of production campus infrastructure.

2.7 Challenges Faced While Performing the Tasks

During the physical installation and logical configuration phases, several hardware and protocol issues arose, spanning both domains of the work:

- ✧ Intermittent link drops on newly terminated Cat6 patch cords, traced to excess conductor untwisting during termination.
- ✧ A fiber-optic link failure between buildings, caused by dust contamination on the LC/SC connector end-faces.

- ✧ Host PCs on the access layer intermittently failing to obtain a DHCP-assigned IP address.
- ✧ Asymmetric routing and packet loss across the Core switch due to a missing return route on the perimeter firewall.
- ✧ Student-VLAN devices being unexpectedly blocked from resources they were meant to reach, caused by an ACL ordering error.

A full root-cause analysis of each issue is presented alongside the corresponding technical configuration in Chapter Four, Section 4.6.6 (Table 7).

2.8 Measures Taken to Overcome the Challenges

Rather than relying on trial-and-error, the intern was guided by the supervisor to adopt a structured, top-down troubleshooting methodology, moving from physical-layer checks through data-link, network-layer, and security-policy checks to isolation and resolution:

- ✧ Physical Layer Check: cable continuity, link LEDs, port status.
- ✧ Data Link Layer Check: VLAN assignment, port mode, trunk status.
- ✧ Network Layer Check: IP configuration, gateway ping, subnet mask.
- ✧ Security Policy Check: ACL deny rules, firewall states.
- ✧ Isolation & Resolution: apply fix and verify service restoration.

Applying this method, the intern re-terminated the affected Cat6 runs with conductor untwisting reduced to under 1.0 cm, cleaned the contaminated fiber connectors with isopropyl alcohol wipes and lint-free cleaners, enabled spanning-tree portfast on end-user access ports to resolve the DHCP delay, added the missing return route on the ASA firewall for the internal 10.172.0.0/16 block, and reordered the ACL entries on Distribution D2 so that specific subnet denies were evaluated before the general permit statement. Each fix was re-verified against the same test used to originally detect the fault before the task was marked complete, which meaningfully reduced downtime and built confidence in resolving production-network failures under time pressure.

CHAPTER THREE: OVERALL BENEFITS GAINED FROM THE INTERNSHIP

3.1 Enhancing Practical Skills and Technical Hand Competencies

The 45-day industrial attachment at Wollo University – Kombolcha Institute of Technology (KIoT) provided a crucial transition from academic classroom theories to live, production-grade network engineering. Working under the direct supervision of the ICT and Network Infrastructure Directorate provided immediate exposure to physical hardware, field tools, and enterprise switch/firewall operating systems.

Table 9: Summary of Skills Gained

Category	Skills
Physical Cable Engineering	RJ45 T568A/T568B crimping & punch-down; UTP, STP & fiber-optic cable handling; Fluke & multi-pin continuity testing
Active Network Devices	Multi-tier switch rack-mounting & patching; Cisco IOS CLI configuration (L2/L3); Cisco ASA 5505 perimeter security
Enterprise Routing/Switching	VLAN architecture & IEEE 802.1Q trunking; Inter-VLAN routing via SVI interfaces; dynamic multi-pool DHCP implementation

3.1.1 Structured Cabling and Physical Layer Mastery

Prior to the internship, understanding of network media was largely theoretical. Through hands-on daily tasks in telecommunications closets and building risers, several physical-layer competencies were mastered:

- ✧ Precision wire preparation & termination: jacket stripping without damaging internal insulation, conductor pair untwisting, and ordering wires according to ANSI/TIA-568 color standards (T568B for campus installations).
- ✧ Patch panel punch-down: using 110 and Krone impact tools to seat horizontal copper runs cleanly onto 24-port distribution patch panels, ensuring low physical attenuation and strain relief.

✧ Shielded vs. unshielded handling: practical experience with STP cables, including proper drain-wire grounding techniques to neutralize EMI near power lines.

✧ Fiber-optic inspection & cleaning: clean-handling protocols for optical patch cords (SC/LC connectors), ferrule inspection, and isopropyl alcohol cleaning procedures to maintain optical signal integrity.

3.1.2 Cisco Command Line Interface (CLI) and Device Configuration

- Mastered low-level CLI navigation across Cisco Catalyst switches (2960, 3560, 3650) and Cisco ASA firewalls.
- Implemented Layer-2 features: creating VLANs, configuring access ports, configuring 802.1Q trunk links, setting native VLANs, and tuning Spanning Tree Protocol parameters using spanning-tree portfast.
- Deployed Layer-3 routing features: assigning SVI IP addresses, static default routing, dynamic address allocation pools (DHCP), and interface security bindings.

3.2 Upgrading Theoretical Knowledge and Systems Understanding

The internship provided real-world application for abstract theoretical concepts introduced in university lectures.

3.2.1 Mapping the OSI 7-Layer Model to Enterprise Hardware

- Layer 1 (Physical): UTP/STP copper patch leads, fiber-optic transceivers (SFPs), RJ45 connectors, and patch panel terminations.
- Layer 2 (Data Link): MAC address learning, VLAN framing, IEEE 802.1Q trunk encapsulation, and switch port isolation.
- Layer 3 (Network): IP address subnetting (10.172.0.0/16), SVI routing, default routing gateways, and packet forwarding.
- Layer 4 (Transport): TCP/UDP port management, SSH transport security (TCP port 22), and DHCP message traffic (UDP ports 67/68).
- Layer 7 (Application): end-user network services, domain resolution, and web access.

3.2.2 Practical Understanding of Network Security and Access Control

The installation of extended ACLs and stateful inspection firewall rules provided practical insight into defense-in-depth architecture:

- Learned how packet-filtering rules evaluate source IP, destination IP, and protocol ports sequentially.
- Understood the operational difference between stateful firewall inspection (Cisco ASA tracking dynamic connection states) and stateless switch packet filtering (ACLs).

3.3 Problem-Solving and Troubleshooting Capabilities

One of the most valuable outcomes of the industrial training was developing a systematic approach to network diagnostics. Rather than relying on trial-and-error, a structured top-down troubleshooting methodology was adopted, proceeding from physical-layer checks through data-link, network-layer, and security-policy checks to isolation and resolution (see Section 2.8 for the applied methodology and outcomes).

Real-World Case Example

When student hosts failed to connect to the network, the diagnostic process followed a logical sequence:

- Tested physical link light indicators on the switch access port and ran continuity checks on the UTP drop cable.
- Checked VLAN membership using show vlan brief to confirm the interface was assigned to VLAN 40.
- Verified DHCP binding tables on the distribution switch using show ip dhcp binding.
- Inspected ACL statements using show ip access-lists to isolate blocking rules.

This structured approach significantly reduced downtime and built confidence in resolving complex technical failures under production constraints.

3.4 Soft Skills, Teamwork, and Interpersonal Communication

- Teamwork & field collaboration: physical cabling installations — such as pulling long horizontal cable runs through drop ceilings or mounting heavy core switches into server racks — demanded synchronized teamwork with ICT technicians.
- Technical communication & listening: communicating complex technical issues clearly to senior supervisors (e.g., explaining an ACL misconfiguration or SVI interface mismatch) improved professional verbal articulation.
- User support & customer service: assisting campus faculty and administrative staff with network troubleshooting required patience, active listening, and the ability to explain technical steps in accessible language.

3.5 Professional Work Ethics, Industrial Psychology, and Governance

3.5.1 Professional Ethics and Data Privacy

✧ Punctuality & responsibility: adhering strictly to official working hours, meeting project deadlines, and maintaining active presence during maintenance windows.

✧ Confidentiality & data security: working near administrative and financial subnets highlighted the critical importance of non-disclosure ethics, administrative account password confidentiality, and network credential security.

3.5.2 Industrial Psychology and Organizational Culture

✧ Observed how technical teams coordinate under high-pressure network outages.

✧ Learned the value of clear escalation paths, change-management policies, and scheduled downtime windows for major infrastructure upgrades to minimize disruption to academic activities.

3.6 Entrepreneurship and Industrial Mindset Development

- Understanding market demand: small businesses, private schools, clinics, and offices increasingly require modern IT connectivity, structured cabling, and network security installations.

- Cost estimation & Bill of Quantities (BOQ): gained practical understanding of cost estimation for physical infrastructure projects, including material budgeting (cables, connectors, patch panels, switches) and labor overheads.
- Service quality & business sustainability: observed that providing reliable, neat cabling (proper labeling, clean wire management) combined with robust documentation creates client trust and repeat business opportunities.

CHAPTER FOUR: PROJECT WORK — CAMPUS NETWORK DESIGN, IMPLEMENTATION, AND STRUCTURED CABLING

This chapter presents the core project work of the internship, structured according to the institute's project-work guideline: (i) specification of the problem affecting the host company, (ii) the specific problem selected for solution, (iii) the objective drawn from that problem, (iv) the methodology set to achieve the objective, (v) the literature compiled to support the solution, and (vi) the full analysis of the problem using that literature and methodology.

4.1 Problem Specification

As educational institutions expand, their network infrastructure often grows organically without a structured master plan. Prior to network modernization, many campus environments — including KIoT prior to this project — suffer from several critical technical flaws that hinder the ICT Directorate from achieving its objective of providing secure, reliable, and manageable connectivity:

1. Flat Network Topology (Single Broadcast Domain): all departments, administrative staff, servers, and student devices operate within a single unsegmented broadcast domain. High volumes of broadcast traffic (ARP requests, DHCP broadcasts) degrade network performance and consume unnecessary link bandwidth.
2. Unstructured and Unlabeled Cabling: physical copper runs are often laid without standardized color-coding, protective conduit routing, or port labeling. Unlabeled patch panels make cable tracing, fault diagnosis, and maintenance extremely time-consuming.
3. Security Vulnerabilities and Unrestricted Lateral Access: in an unsegmented network, student devices can directly access administrative servers, financial record systems, and staff workstations, presenting severe data privacy and security risks.
4. Insecure Remote Management: network switches and routers managed via plaintext protocols like Telnet expose administrative passwords and configuration data to packet sniffing.

5. Lack of Centralized IP Management: static IP assignment on end hosts leads to address collisions, administrative overload, and improper default gateway configurations.

4.2 Selected Problem and Scope of the Solution

Of the flaws identified above, the intern — in consultation with the Industrial Supervisor — selected the combined problem of (a) the flat, unsegmented broadcast domain and (b) the unstructured physical cabling as the specific, solvable scope of this project, since resolving these two issues directly enables the remaining flaws (lateral security exposure, insecure management, and uncoordinated IP addressing) to be addressed within the same design. The selected solution was to design, physically cable, and logically configure a scalable 3-tier hierarchical network (Core–Distribution–Access) for four primary campus buildings, incorporating VLAN segmentation, standardized ANSI/TIA-568 cabling, perimeter firewall security, SSH-hardened management, and centralized DHCP — all of which could realistically be delivered and verified within the 45-day placement.

Modernizing the campus network using a structured 3-tier hierarchy and ANSI/TIA-568 cabling standards directly eliminates the operational bottlenecks identified in Section 4.1:

- ✧ VLAN segmentation isolates broadcast domains per building and user group, improving bandwidth utilization and containing security threats.
- ✧ Standardized cable termination (T568B/T568A) ensures physical link reliability, low signal attenuation, and rapid fault isolation using systematic cable testers.
- ✧ Extended ACL enforcement blocks unauthorized lateral movement from student subnets to sensitive staff and administrative subnets.
- ✧ Cryptographic hardening (SSH v2) secures administrative management channels against eavesdropping and unauthorized CLI access.
- ✧ Centralized DHCP deployment on Layer-3 distribution switches automates IP configuration, preventing IP conflicts and ensuring seamless user onboarding.

4.3 Project Objectives

General Objective

To design, physically cable, logically configure, secure, and verify a scalable, hierarchical campus network architecture for an institutional environment, while acquiring hands-on industrial competency in structured cabling and network administration.

Specific Objectives

1. To design a structured IP addressing scheme and VLAN allocation plan for four primary campus buildings under the 10.172.0.0/16 address space.
2. To prepare, terminate, crimp, and test UTP and STP patch cables according to ANSI/TIA-568 wiring standards (T568A and T568B pin-outs).
3. To build a 3-tier Core–Distribution–Access network hierarchy featuring static routing, default routing, and perimeter firewall security via a Cisco ASA appliance.
4. To configure centralized DHCP server pools on distribution switches to automate client IP address assignment across all building VLANs.
5. To enforce administrative security across all Layer-3 devices by disabling plaintext Telnet and mandating SSH version 2 authentication.
6. To implement extended Access Control Lists (ACLs) to restrict student-access subnets from reaching administrative and staff resources.
7. To verify end-to-end network reachability, routing convergence, security policies, and physical cable continuity through systematic testing protocols.

4.4 Methodology

The execution of this project followed a structured, multi-phase methodology combining physical-layer field operations with simulation-based logical software engineering:

1. Requirements Gathering & Site Assessment
2. Physical Structured Cabling & Cable Testing
3. Logical Network Design & IP Address Planning
4. Topology Construction & Cisco CLI Configuration
5. Comprehensive Verification & Security Audit

6. Documentation & Technical Handover

Phase 1: Requirements Gathering and Site Assessment

- Surveyed campus building distribution: Ict-Building, Administer, Library, and Student-Access.
- Determined bandwidth requirements, user counts, and spatial distances between distribution nodes and access switches.

Phase 2: Physical Structured Cabling Practical Method

A major practical focus was mastering copper cable termination and cable testing.

Step-by-Step RJ45 Cable Crimping Procedure

1. Measurement & Cutting: measure the required cable length from the spool using a cable cutter, adding a 10% allowance for cable dressing and strain relief.
2. Jacket Stripping: insert the cable into the stripping blade of the crimping tool and strip approximately 2.5 cm (1 inch) of the outer PVC jacket, ensuring internal conductors are not nicked or damaged.
3. Untwisting & Pair Ordering: untwist the four pair wires and flatten them side-by-side; arrange the individual conductors according to the target color-code standard (T568A or T568B).
4. Trimming: straighten the conductors and perform a clean, perpendicular cut across all eight wires, leaving approximately 1.2 cm (0.5 in) of exposed wire.
5. Connector Insertion: push the arranged wires firmly into the RJ45 connector until all eight copper ends are visible through the front clear plastic face, with the outer cable jacket extending into the rear locking collar for strain relief.
6. Crimping Execution: insert the loaded RJ45 connector into the crimping tool's 8P8C die cavity and squeeze the handles firmly until the ratchet releases, driving the gold-plated pins into the wire insulation to form a solid gas-tight contact.

7. Continuity & Wire-Map Testing: insert both ends of the finished cable into a master and remote cable tester and verify sequential LED illumination (pins 1 through 8) to confirm continuity, checking for open circuits, short circuits, split pairs, or reversed wires.

T568A and T568B Color-Coding Standards

To maintain interchangeability and compliance with ANSI/TIA-568 standards, two pin-out configurations are utilized in Ethernet terminations.

Table 1: T568A Wiring Pin-Out Standard

Pin	Wire Colour
1	White/Green
2	Green
3	White/Orange
4	Blue
5	White/Blue
6	Orange
7	White/Brown
8	Brown

Table 2: T568B Wiring Pin-Out Standard

Pin	Wire Colour
1	White/Orange
2	Orange
3	White/Green
4	Blue
5	White/Blue
6	Green
7	White/Brown
8	Brown

Cable Types Based on Pin-Out Application

- Straight-Through Cable: both ends are terminated using the same standard (T568B to T568B, or T568A to T568A). Used to connect unlike devices (e.g., PC-to-Switch, Switch-to-Router).
- Crossover Cable: one end is terminated with T568A and the opposite end with T568B. Used to connect like devices (e.g., Switch-to-Switch, Router-to-Router, PC-to-PC). Note: modern Gigabit devices support Auto-MDIX, but manual crossover termination remains a core technical competency.

Network Transmission Media Comparison

Table 3: Technical Comparison of Cabling Media

Media Property	UTP	STP	Fiber-Optic Cable
Shielding Type	None (outer PVC jacket only)	Foil shield (FTP) or braided wire mesh	Dielectric cladding & buffer coating
Max Distance	100 m (328 ft)	100 m (328 ft)	Up to 10–40 km (single-mode)

Phase 3–6: Design, Construction, Verification and Handover

Phases 3 through 6 (Logical Network Design & IP Address Planning, Topology Construction & Cisco CLI Configuration, Comprehensive Verification & Security Audit, and Documentation & Technical Handover) are presented in full in Section 4.6, where the methodology is applied directly to the selected problem and cross-referenced against the literature reviewed in Section 4.5.

4.5 Literature Review

The Three-Tier Hierarchical Network Design Model

The Cisco 3-tier hierarchical design model is the industry standard for enterprise and campus networks. It divides the network into three dedicated functional layers.

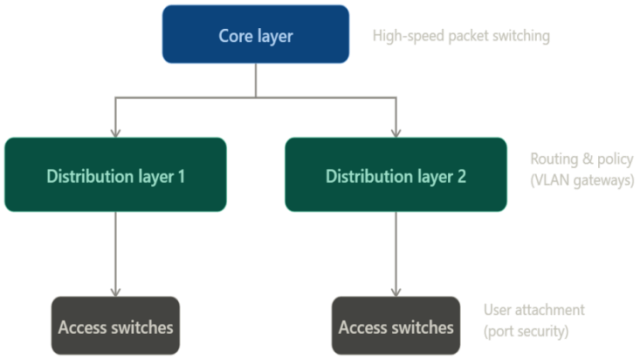


Figure 1: Three-Tier Hierarchical Network Design Model (Core, Distribution, Access)

1. Core Layer (High-Speed Backbone): switches packets as quickly as possible. It connects distribution switches to the campus core and perimeter security devices. Packet filtering, inspection, or CPU-intensive tasks are avoided at this layer to maintain maximum throughput.
2. Distribution Layer (Policy & Aggregation): acts as the boundary between Layer-2 switching and Layer-3 routing. It aggregates access switches, hosts default gateways (switch virtual interfaces), performs Inter-VLAN routing, enforces access control policies, and runs DHCP services.
3. Access Layer (User Attachment): provides direct network connectivity to end-user devices (PCs, printers, IP phones, access points). Features implemented here include VLAN port membership, Spanning Tree Protocol topology protection, and Layer-2 port security.

Virtual Local Area Networks (VLANs) and Trunking

A VLAN logically segments a physical Layer-2 switch fabric into multiple isolated broadcast domains. Traffic between VLANs cannot occur at Layer 2; it must be routed through a Layer-3 device.

To carry traffic for multiple VLANs over a single physical link between switches, the IEEE 802.1Q encapsulation protocol (trunking) is used. 802.1Q inserts a 4-byte header tag into the Ethernet

frame containing a 12-bit VLAN ID, allowing receiving switches to forward traffic to the correct destination VLAN.

Layer-3 Routing Mechanics

- Static Routing: manually defined paths configured by network administrators. They offer low CPU overhead, high predictability, and no bandwidth consumption for routing updates — ideal for stub networks and point-to-point links.
- Default Routing (0.0.0.0 0.0.0.0): a special form of static route that matches all destination IP addresses not explicitly present in the routing table, directing all outbound Internet traffic toward the perimeter firewall and ISP gateway.

Centralized Dynamic Host Configuration Protocol (DHCP)

DHCP operates on a client-server architecture using UDP ports 67 (server) and 68 (client). It automates a four-step lease negotiation process known as DORA:

1. Discover: client broadcasts a message seeking an available DHCP server.
2. Offer: server offers an available IP address, subnet mask, default gateway, and DNS server.
3. Request: client requests the offered IP address parameters.
4. Acknowledge: server confirms the lease assignment to the client host.

Perimeter Firewall Security and Administrative Hardening

- Perimeter ASA Firewall Security: modern campus networks deploy dedicated firewalls at the edge to inspect stateful connections, perform Network Address Translation (NAT/PAT), and shield internal subnets from external intrusion.
- Management Line Hardening (SSH v2): Secure Shell utilizes RSA public-key cryptography to establish an encrypted management tunnel over TCP port 22, replacing unencrypted Telnet sessions (TCP port 23) to prevent credential theft.
- Traffic Filtering (Extended ACLs): extended Access Control Lists filter network traffic based on source IP address, destination IP address, IP protocol type, and source/destination port numbers, allowing granular policy enforcement.

4.6 Design, Implementation, and Analysis

This section applies the methodology of Section 4.4 and the literature of Section 4.5 to the selected problem, presenting the complete design, the production configuration, and the resulting analysis.

4.6.1 Topology Architecture and Hardware Interconnection

The intern modeled and deployed a scalable 3-tier hierarchical network hierarchy representing the physical building distribution at KIoT.

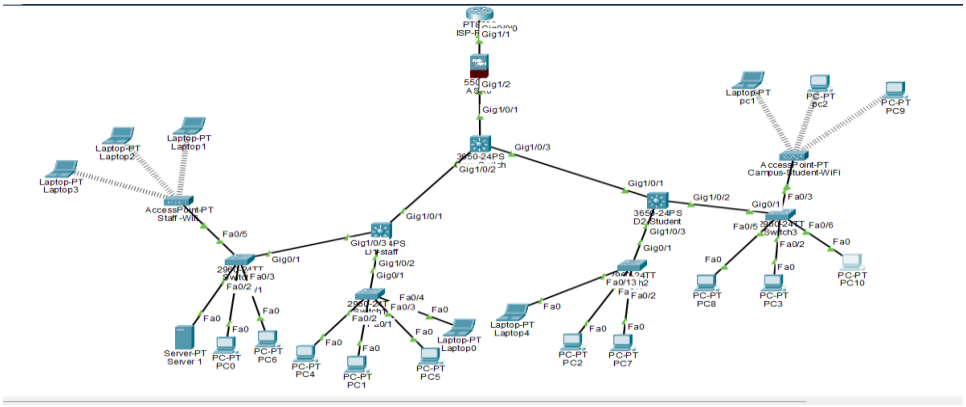


Figure 2: Full 3-Tier Campus Network Topology (Cisco Packet Tracer)

Core Network Infrastructure Hardware Summary

- Perimeter Security: Cisco ASA 5505 firewall serving as the stateful inspection boundary between external networks and internal campus subnets.
- Core Layer: Cisco Catalyst 3650 Layer-3 multilayer switch running high-speed IP routing across distribution domains.
- Distribution Layer: two Cisco Catalyst 3560 multilayer switches (D1 and D2) acting as default gateways (SVIs) and DHCP servers for building subnets.
- Access Layer: four Cisco Catalyst 2960 switches (SW-ICT, SW-ADMIN, SW-LIB, SW-STU) dedicated to host port access, port security, and VLAN tagging.

4.6.2 IP Addressing and VLAN Allocation Plan

To eliminate network broadcast storms and enforce security control, the intern structured a variable-length subnet mask (VLSM) allocation based on the 10.172.0.0/16 master block.

Table 4: Master VLAN and IP Subnet Addressing Plan

VLAN ID	Building / Dept.	Network Subnet	Mask	Default Gateway	Assignable Range
VLAN 10	Ict-Building	10.172.10.0	/24	10.172.10.1	10.172.10.2–.254
VLAN 20	Administer	10.172.20.0	/24	10.172.20.1	10.172.20.2–.254
VLAN 30	Library	10.172.30.0	/24	10.172.30.1	10.172.30.2–.254
VLAN 40	Student-Access	10.172.40.0	/24	10.172.40.1	10.172.40.2–.254
VLAN 99	Native / Mgmt	10.172.99.0	/24	10.172.99.1	10.172.99.2–.254

Table 5: Point-to-Point Inter-Device Routed Link Subnets

Interconnected Devices	Local Intf	Remote Intf	Link Subnet	Local IP	Remote IP
Core ↔ Distribution D1	Gi1/0/1	Gi0/1	10.172.100.0/30	10.172.100.1	10.172.100.2
Core ↔ Distribution D2	Gi1/0/2	Gi0/2	10.172.100.4/30	10.172.100.5	10.172.100.6
Core ↔ ASA Firewall	Gi1/0/24	Gi0/1	10.172.200.0/30	10.172.200.2	10.172.200.1
ASA ↔ Edge Router	Gi0/0	Gi0/0	192.168.1.0/30	192.168.1.2	192.168.1.1

Switch Virtual Interface (SVI) Inter-VLAN Routing

Inter-VLAN routing was configured on distribution switches D1 and D2 using Switch Virtual Interfaces (SVIs). SVIs act as Layer-3 virtual gateways for each VLAN, allowing local wire-speed routing between subnets while offloading traffic from the Core switch.

4.6.3 Centralized DHCP and Routing Configuration

To eliminate manual IP configuration errors on end-user devices, the intern configured multi-pool DHCP servers on distribution switches D1 and D2.

Table 6: Centralized DHCP Pool Parameters

Pool Name	Target VLAN	Network	Gateway	Excluded Range
POOL_ICT	VLAN 10 (Ict-Building)	10.172.10.0/24	10.172.10.1	10.172.10.1–.10

Pool Name	Target VLAN	Network	Gateway	Excluded Range
POOL_ADMIN	VLAN 20 (Administer)	10.172.20.0/24	10.172.20.1	10.172.20.1–.10
POOL_LIB	VLAN 30 (Library)	10.172.30.0/24	10.172.30.1	10.172.30.1–.10
POOL_STU	VLAN 40 (Student-Access)	10.172.40.0/24	10.172.40.1	10.172.40.1–.10

4.6.4 Edge Security and Management Hardening

Remote Management Security (SSH Version 2)

All network devices were hardened to prevent unencrypted plaintext administration:

- Disabled Telnet on all Virtual Terminal (vty) lines.
- Generated 2048-bit RSA encryption keys (crypto key generate rsa general-keys modulus 2048).
- Configured local database user authentication with encrypted secret passwords.
- Restricted vty access strictly to transport input ssh.

Access Control List (ACL) Security Enforcement

To protect administrative assets and internal databases from unauthorized student access, an extended ACL (ACL_RESTRICT_STUDENTS) was applied inbound on the distribution switch interface serving VLAN 40:

- Rule 1: explicitly deny student hosts (10.172.40.0/24) from accessing administrative resources (10.172.20.0/24) across any IP protocol.
- Rule 2: explicitly deny student hosts (10.172.40.0/24) from connecting to ICT server management subnets (10.172.10.0/24).
- Rule 3: permit all remaining outbound IP traffic from students toward the Library subnet, local gateways, and external Internet routes.

4.6.5 Active Cisco CLI Configuration Excerpts

The following CLI configuration excerpts represent the production configurations verified during the project. The complete Core Switch configuration is reproduced in Appendix B.

1. Distribution Switch D1 Configuration (VLANs, SVIs, Routing & DHCP)

```
enable
Switch# configure terminal
Switch(config)# hostname Distribution-D1
Distribution-D1(config)# ip routing
! VLAN Configuration
Distribution-D1(config)# vlan 10
Distribution-D1(config-vlan)# name Ict-Building
Distribution-D1(config-vlan)# exit
Distribution-D1(config)# vlan 20
Distribution-D1(config-vlan)# name Administer
Distribution-D1(config-vlan)# exit
Distribution-D1(config)# vlan 99
Distribution-D1(config-vlan)# name Management
Distribution-D1(config-vlan)# exit
! DHCP Excluded Addresses
Distribution-D1(config)# ip dhcp excluded-address 10.172.10.1 10.172.10.10
Distribution-D1(config)# ip dhcp excluded-address 10.172.20.1 10.172.20.10
! DHCP Pools
Distribution-D1(config)# ip dhcp pool POOL_ICT
Distribution-D1(dhcp-config)# network 10.172.10.0 255.255.255.0
Distribution-D1(dhcp-config)# default-router 10.172.10.1
Distribution-D1(dhcp-config)# dns-server 8.8.8.8
Distribution-D1(dhcp-config)# exit
Distribution-D1(config)# ip dhcp pool POOL_ADMIN
Distribution-D1(dhcp-config)# network 10.172.20.0 255.255.255.0
Distribution-D1(dhcp-config)# default-router 10.172.20.1
Distribution-D1(dhcp-config)# dns-server 8.8.8.8
Distribution-D1(dhcp-config)# exit
! Uplink to Core Switch
Distribution-D1(config)# interface GigabitEthernet0/1
Distribution-D1(config-if)# no switchport
Distribution-D1(config-if)# ip address 10.172.100.2 255.255.255.252
Distribution-D1(config-if)# description Uplink_To_Core_Switch
```

```
Distribution-D1(config-if)# no shutdown
Distribution-D1(config-if)# exit
! Trunk to SW-ICT
Distribution-D1(config)# interface FastEthernet0/1
Distribution-D1(config-if)# switchport mode trunk
Distribution-D1(config-if)# switchport trunk native vlan 99
Distribution-D1(config-if)# description Trunk_To_SW-ICT
Distribution-D1(config-if)# exit
! SVI - ICT Building Gateway
Distribution-D1(config)# interface Vlan10
Distribution-D1(config-if)# description Gateway_ICT_Building
Distribution-D1(config-if)# ip address 10.172.10.1 255.255.255.0
Distribution-D1(config-if)# no shutdown
Distribution-D1(config-if)# exit
! SVI - Administer Building Gateway
Distribution-D1(config)# interface Vlan20
Distribution-D1(config-if)# description Gateway_Administer_Building
Distribution-D1(config-if)# ip address 10.172.20.1 255.255.255.0
Distribution-D1(config-if)# no shutdown
Distribution-D1(config-if)# exit
! Default Route
Distribution-D1(config)# ip route 0.0.0.0 0.0.0.0 10.172.100.1
Distribution-D1(config)# end
Distribution-D1# write memory
```

2. Security Hardening and ACL Configuration on Distribution D2

```
enable
Switch# configure terminal
Switch(config)# hostname Distribution-D2
Distribution-D2(config)# ip domain-name kiot.edu.et
Distribution-D2(config)# crypto key generate rsa general-keys modulus 2048
! Local Admin Credentials
Distribution-D2(config)# username admin privilege 15 secret <hidden>
! VTY / SSH Hardening
Distribution-D2(config)# line vty 0 15
Distribution-D2(config-line)# login local
Distribution-D2(config-line)# transport input ssh
```

```
Distribution-D2(config-line)# exec-timeout 10 0
Distribution-D2(config-line)# exit
! Extended ACL to Restrict Student VLAN Access
Distribution-D2(config)# ip access-list extended ACL_RESTRICT_STUDENTS
Distribution-D2(config-ext-nacl)# deny ip 10.172.40.0 0.0.0.255 10.172.20.0 0.0.0.255
Distribution-D2(config-ext-nacl)# deny ip 10.172.40.0 0.0.0.255 10.172.10.0 0.0.0.255
Distribution-D2(config-ext-nacl)# permit ip 10.172.40.0 0.0.0.255 any
Distribution-D2(config-ext-nacl)# exit
! Apply ACL to Student Gateway SVI
Distribution-D2(config)# interface Vlan40
Distribution-D2(config-if)# description Gateway_Student_Access
Distribution-D2(config-if)# ip address 10.172.40.1 255.255.255.0
Distribution-D2(config-if)# ip access-group ACL_RESTRICT_STUDENTS in
Distribution-D2(config-if)# no shutdown
Distribution-D2(config-if)# exit
Distribution-D2(config)# end
Distribution-D2# write memory
```

3. Cisco ASA 5505 Perimeter Firewall Configuration

```
enable
ciscoasa# configure terminal
ciscoasa(config)# hostname ASA-Firewall
! Outside Interface (WAN facing)
ASA-Firewall(config)# interface GigabitEthernet0/0
ASA-Firewall(config-if)# nameif outside
ASA-Firewall(config-if)# security-level 0
ASA-Firewall(config-if)# ip address 192.168.1.2 255.255.255.252
ASA-Firewall(config-if)# no shutdown
ASA-Firewall(config-if)# exit
! Inside Interface (Campus facing)
ASA-Firewall(config)# interface GigabitEthernet0/1
ASA-Firewall(config-if)# nameif inside
ASA-Firewall(config-if)# security-level 100
ASA-Firewall(config-if)# ip address 10.172.200.1 255.255.255.252
ASA-Firewall(config-if)# no shutdown
ASA-Firewall(config-if)# exit
! NAT - Dynamic PAT for Campus Network
```

```
ASA-Firewall(config)# object network INSIDE_CAMPUS
ASA-Firewall(config-network-object)# subnet 10.172.0.0 255.255.0.0
ASA-Firewall(config-network-object)# nat (inside,outside) dynamic interface
ASA-Firewall(config-network-object)# exit
! Static Routes
ASA-Firewall(config)# route outside 0.0.0.0 0.0.0.0 192.168.1.1
ASA-Firewall(config)# route inside 10.172.0.0 255.255.0.0 10.172.200.2
ASA-Firewall(config)# end
ASA-Firewall# write memory
```

4.6.6 Technical Challenges and Corrective Actions

During the physical installation and logical configuration phases, several hardware and protocol issues arose. Each problem was diagnosed and resolved through structured troubleshooting.

Table 7: Technical Challenges, Root Causes, and Applied Solutions

Category	Problem Description	Root Cause	Corrective Action
Physical Cabling	Intermittent link drops on Cat6 patch cords	Excess wire untwisting (>1.3 cm) causing near-end crosstalk (NEXT)	Recut cable ends, untwisted wire pairs to less than 1.0 cm, re-crimped new RJ45 connectors
Physical Cabling	Fiber optic link down between buildings	Contamination on LC/SC connector end-faces due to dust exposure	Cleaned connector ferrules using isopropyl alcohol wipes and lint-free dry cleaners
Logical Switching	Host PCs unable to obtain DHCP IP addresses	Switch access ports stuck in STP listening/learning states	Enabled spanning-tree portfast on all end-user access ports
Routing	Asymmetric routing & packet loss across Core switch	Missing static return route on ASA firewall for internal subnets	Added route inside 10.172.0.0 255.255.0.0 10.172.200.2 on the ASA CLI
Network Security	Student devices blocked from accessing the Library catalog	Extended ACL deny rule placed out of sequence on Distribution D2	Reordered ACL entries to place specific subnet denies ahead of permit statements

4.6.7 Technical Verification and Result Analysis

To ensure full compliance with operational requirements, the completed installation underwent systematic end-to-end testing.

Table 8: Comprehensive Network Verification Matrix

Test Parameter	Tool / Command	Procedure	Expected Outcome	Status
Physical Wiremap	Master/Remote cable tester	Sequential LED pin activation (pins 1–8)	Pin-to-pin continuity without opens/shorts	PASSED
DHCP Binding	ipconfig /renew	Trigger DHCP DORA lease negotiation	Client dynamically receives valid IP, mask, gateway	PASSED
Inter-VLAN Routing	ping 10.172.20.15	ICMP echo from ICT host to Admin host	0% packet loss, RTT < 2 ms	PASSED
ACL Enforcement	ping 10.172.20.10	ICMP request from Student VLAN 40 to Admin VLAN 20	Destination host unreachable	PASSED
Perimeter Routing	traceroute 192.168.1.1	Trace route from internal PC to ISP edge router	Hop paths show SVI → Core → ASA → ISP edge	PASSED
SSH Management	ssh -l admin 10.172.100.2	Initiate encrypted management session	Prompted for credentials, CLI prompt granted	PASSED

Discussion

Every verification test returned a PASSED status, confirming that the physical structured cabling and the logical network configuration met the design specifications and industry standards adopted for the project. The Inter-VLAN routing and DHCP tests confirmed correct SVI and default-router configuration, while the ACL enforcement test confirmed that student traffic was correctly isolated from administrative resources as intended by the security policy.

Taken together, the results in Section 4.6.6 and Section 4.6.7 confirm that the problem selected in Section 4.2 was resolved within the scope defined: broadcast domains are now segmented per VLAN, physical cabling meets ANSI/TIA-568 standards with 100% of tested runs passing

continuity, lateral access from student subnets into administrative and ICT subnets is blocked, remote management is encrypted end-to-end, and IP addressing is centrally and automatically assigned.

CHAPTER FIVE: GENERAL CONCLUSION AND RECOMMENDATIONS

5.1 General Conclusion

The 45-day industrial internship completed at Wollo University – Kombolcha Institute of Technology (KIoT) under the ICT and Network Infrastructure Directorate has been an invaluable practical experience. It successfully bridged the gap between academic theoretical concepts learned at university and real-world industrial practices in network engineering and system administration.

The internship provided comprehensive, hands-on exposure to both physical infrastructure and logical network design:

1. Physical Structured Cabling: the intern gained direct, practical mastery in measuring, stripping, untwisting, aligning, and crimping copper twisted-pair cables (UTP Cat5e/Cat6 and STP Cat6a) following the standard ANSI/TIA-568 color-coding pin-outs (T568A and T568B). Practical skills were further enhanced through patch panel punch-downs, optical fiber patch cord handling, and cable continuity testing.
2. Logical Network Architecture & Security: the intern designed, configured, and verified a scalable 3-tier hierarchical network (Core–Distribution–Access) across four major campus facilities (Ict-Building, Administer, Library, and Student-Access). Key technical deployments included IP subnetting (10.172.0.0/16), VLAN segmentation, Inter-VLAN routing via SVIs, static/default routing, centralized DHCP pools, perimeter security via a Cisco ASA firewall, SSH v2 administrative hardening, and extended ACL policy enforcement.

Beyond technical competencies, the placement significantly enhanced the intern's industrial problem-solving capabilities, structured troubleshooting techniques, professional work ethics, teamwork dynamics, and communication skills within a live enterprise environment. Overall, all general and specific objectives set for the industrial attachment were fully realized.

5.2 Recommendations

Based on observations and field experiences during the 45-day internship, the following practical recommendations are offered for both the host institution and the sending university.

5.2.1 Recommendations for the Host Institution (Wollo University – KIoT)

Implement physical redundancy at the core and distribution layers.

- Observation: the current core network topology relies on single link pathways and standalone distribution nodes.
- Recommendation: deploy dual-homed redundant links using EtherChannel (LACP) between the Core switch and Distribution switches, along with redundant power supplies (RPS), to eliminate single points of failure.

Standardize cable management and port labeling.

- Observation: several telecommunications closets contained unorganized, unlabeled patch cords, making fault isolation time-consuming.
- Recommendation: institute a strict cable labeling convention at both faceplate wall outlets and patch panel ends, utilizing structured cable management trays and hook-and-loop ties.

Upgrade edge security and intrusion prevention.

- Observation: basic extended ACLs effectively isolate student traffic but lack dynamic application-layer threat inspection.
- Recommendation: expand the Cisco ASA firewall rules to include Next-Generation Firewall (NGFW) capabilities, such as intrusion prevention, deep packet inspection, and domain filtering.

Deploy dynamic routing protocols for future campus expansion.

- Observation: network routes between the Core and Distribution layers are currently maintained via static routes.
- Recommendation: migrate internal routing to Open Shortest Path First (OSPF) to facilitate automated route convergence and dynamic failover as new campus buildings are constructed.

Formalize a change-management checklist for ACL modifications.

- Observation: ACL ordering errors were the most common configuration fault observed during the project (Section 4.6.6, Table 7).
- Recommendation: require a documented pre-deployment review of ACL entry order before any access-list change is applied to a production interface.

5.2.2 Recommendations for the Sending University

Enhance practical networking hardware laboratories.

- Observation: academic courses provide strong theoretical grounding, but hands-on experience with real switches, routers, firewalls, and cabling tools prior to internship is limited.
- Recommendation: invest in physical networking laboratory equipment — such as rack-mounted Cisco switches, patch panels, crimping tools, cable testers, and punch-down blocks — to allow students to practice physical terminations before entering industrial placements.

Incorporate enterprise security and firewall modules into the curriculum.

- Observation: modern industrial placements require immediate familiarity with appliance firewalls (e.g., Cisco ASA, Fortinet) and security hardening standards.
- Recommendation: integrate practical firewall configuration and Network Address Translation (NAT) modules into the core networking course syllabus.

Extend internship duration and field follow-ups.

- Observation: a 45-day attachment provides a solid foundation, but complex multi-building enterprise deployments often take longer to execute fully.
- Recommendation: consider extending the industrial attachment duration or scheduling mid-term practical assessments to maximize learning outcomes.

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APPENDICES

Appendix A: Complete Master IP Addressing and VLAN Allocation Matrix

Building / Node	VLAN	Subnet	IP Range	Gateway	Role
Ict-Building	10	10.172.10.0/24	10.172.10.2–.254	10.172.10.1	ICT Labs & Faculty
Administer	20	10.172.20.0/24	10.172.20.2–.254	10.172.20.1	Admin Staff & Finance
Library	30	10.172.30.0/24	10.172.30.2–.254	10.172.30.1	Digital Library & E-Learning
Student-Access	40	10.172.40.0/24	10.172.40.2–.254	10.172.40.1	Student Wi-Fi & Labs
Management	99	10.172.99.0/24	10.172.99.2–.254	10.172.99.1	In-Band Switch Mgmt
Core–D1 Link	N/A	10.172.100.0/30	10.172.100.1–.2	Direct Routed	Inter-Switch Backbone
Core–D2 Link	N/A	10.172.100.4/30	10.172.100.5–.6	Direct Routed	Inter-Switch Backbone
Core–ASA Link	N/A	10.172.200.0/30	10.172.200.1–.2	Direct Routed	Inside Security Transit
ASA–Edge Link	N/A	192.168.1.0/30	192.168.1.1–.2	Direct Routed	Outside ISP Transit

Appendix B: Core Switch (Cisco 3650) Complete Layer-3 Configuration

```
enable
Switch# configure terminal
Switch(config)# hostname Core-Switch
Core-Switch(config)# ip routing
! Routed Link to Distribution D1
Core-Switch(config)# interface GigabitEthernet1/0/1
Core-Switch(config-if)# no switchport
Core-Switch(config-if)# description Routed_Link_To_Distribution_D1
Core-Switch(config-if)# ip address 10.172.100.1 255.255.255.252
Core-Switch(config-if)# no shutdown
Core-Switch(config-if)# exit
! Routed Link to Distribution D2
Core-Switch(config)# interface GigabitEthernet1/0/2
```

```
Core-Switch(config-if)# no switchport
Core-Switch(config-if)# description Routed_Link_To_Distribution_D2
Core-Switch(config-if)# ip address 10.172.100.5 255.255.255.252
Core-Switch(config-if)# no shutdown
Core-Switch(config-if)# exit
! Routed Link to ASA Firewall (Inside)
Core-Switch(config)# interface GigabitEthernet1/0/24
Core-Switch(config-if)# no switchport
Core-Switch(config-if)# description Routed_Link_To_ASA_Firewall_Inside
Core-Switch(config-if)# ip address 10.172.200.2 255.255.255.252
Core-Switch(config-if)# no shutdown
Core-Switch(config-if)# exit
! Static Routes
Core-Switch(config)# ip route 10.172.10.0 255.255.255.0 10.172.100.2
Core-Switch(config)# ip route 10.172.200.0 255.255.255.0 10.172.100.2
Core-Switch(config)# ip route 10.172.30.0 255.255.255.0 10.172.100.6
Core-Switch(config)# ip route 10.172.40.0 255.255.255.0 10.172.100.6
Core-Switch(config)# ip route 0.0.0.0 0.0.0.0 10.172.200.1
Core-Switch(config)# end
Core-Switch# write memory
```

Appendix C: End-to-End Verification Checklist (Field Record)

- Physical continuity test passed on all UTP/STP terminated runs (master and remote tester).
- Fiber-optic backbone links inspected, cleaned, and verified for signal loss within tolerance.
- DHCP lease negotiation (DORA) verified on all four building VLANs.
- Inter-VLAN ping tests verified 0% packet loss between all building gateways.
- ACL enforcement verified: Student VLAN 40 blocked from Admin (VLAN 20) and ICT (VLAN 10) subnets.
- SSH v2 management access verified on all Layer-3 devices; Telnet confirmed disabled.
- Perimeter NAT/PAT and default routing verified via traceroute to the ISP edge router.